

EDUCATION

Dec 2026 Pittsburgh, PA	Carnegie Mellon University Master of Science in NLP/ML, Language Technologies Institute, School of Computer Science <i>Current Coursework: Advanced Natural Language Processing, Generative AI, Machine Learning</i>
July 2025 Pilani, India	Birla Institute of Technology and Science, Pilani B.E. in Computer Science (CGPA: 9.91/10 , Institute <i>Silver</i> Medalist)

PUBLICATIONS [ALL FIRST/CO-FIRST AUTHOR]

ICCV'25, *ACL	CAIRE: Cultural Attribution of Images by Retrieval-Augmented Evaluation. CEGIS @ ICCV'25, ACL Rolling Review (Accept, to be submitted to *ACL 2026) Paper
LREC-COLING'24	BERT-based Idiom Identification using Language Translation and Word Cohesion. Multiword Expressions and Universal Dependencies @ LREC-COLING'24 Paper
IEEE IS'24	Interpretable Feature Optimization for Sadness Recognition in Speech Emotion Analysis. IEEE 12th International Conference on Intelligent Systems (IS) Paper

EXPERIENCE

Carnegie Mellon University Research Intern (Undergraduate Thesis), NeuLab Advisor: Prof. Graham Neubig Code	Pittsburgh, PA May 2024 – Mar 2025
- Developed a novel metric to quantify cultural relevance of real and generated images, and built an efficient large-scale (6 million entities) text-disambiguation image retrieval system using FAISS, surpassing SOTA LVLMs on the FOCI benchmark. - Augmented LLMs with retrieved cultural context and Chain-of-thought prompting to compute relevance across cultural proxies, achieving +28% F1 on a challenging hand-curated validation set. Achieved Pearson $r > 0.65$ vs human annotations on a dataset comprising universal concepts. Accepted @ ICCV-W & ARR Accept (recommended for *ACL).	
Nanyang Technological University Research Intern, SpeechLab Advisor: Prof. Chng Eng Siong Code	Singapore Mar 2024 – Sep 2024
Fine-tuned LLaMA-3.1-8B with LoRA on the DAIC-WOZ dataset for text-based depression detection, achieving a +7.1% F1 improvement over prior work. Designed a PHQ-8-guided prompting strategy, enhancing both accuracy & interpretability.	
Amazon, Applied Science Summer Intern Advisor: Jitenkumar Rana Code	Bangalore, India May 2023 – Aug 2023
Developed and validated a product brand and model knowledge base using BERT-based NER, applied data augmentation with GPT and Falcon 7B, and created regression-based metrics to detect anomalies in shipping costs.	
BITS Pilani Research Assistant: Led 4 Machine Learning Research Projects	India July 2023 – May 2025
BERT-based Idiom Detection: Designed custom loss functions to improve token-level idiom recognition. Code Interpretable SER: Metaheuristic feature selection for emotion detection; SOTA F1 across 4 popular datasets Code Malware Detection: GNN/sequence models for multi-class classification on imbalanced & obfuscated datasets Code In-Context-Learning with Information Retrieval: Critically evaluated the methodology of the ECIR Best Paper, identifying flaws and proposing corrections, achieving improved performance on downstream NLP classification tasks.	

PROJECTS

* Basic PASCAL Compiler Code	Jan 2024 – May 2024
Implemented a simplified Pascal compiler with LEX/YACC: lexer, parser, semantic checks and intermediate code generation.	
* Show Chain (Blockchain ticketing) Code	Feb 2023 – Apr 2023
Built a blockchain-based movie-ticket distribution system in Java with ZK-proofs for secure transparent transactions.	
* GO MART (E-commerce)	Sep 2022 – Dec 2022
Contributed to a SpringBoot + Vue.js/Tailwind e-commerce app using OOP principles and RESTful design.	

SKILLS

Programming & OS: Python, C/C++, Java, SQL, Linux, High Performance Computing Clusters (HPC)
Libraries and Frameworks: PyTorch, TensorFlow, Numpy, Pandas, Scikit-Learn, HuggingFace, Matplotlib, spaCy
ML: Natural Language Processing, Diffusion Models, Information Retrieval, Computer Vision, Multimodal ML, GNNs